

C1Q/SPP1 Macrophage Spatial Context as a
Reproducible Traceability Feature in Non-Small
Cell Lung Cancer: A Public Multi-Cohort
Cancer-Informatics Evidence Synthesis

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Abstract

Objectives: Public cancer-informatics resources provide an opportunity to test whether macrophage-associated complement programs are reproducible markers of non-small cell lung cancer (NSCLC) tumor-microenvironment context. However, retrospective multi-omics analyses are vulnerable to overinterpretation when prognostic, spatial, and immunotherapy-response signals are presented as clinical tools or mechanistic proof.

Methods: We performed a conservative public-data evidence synthesis centered on the C1Q/SPP1 macrophage axis. Bulk transcriptomic and clinical data from TCGA NSCLC were used for discovery and model-behavior assessment after quality control. External prognostic traceability was evaluated in GSE31210 patients with complete follow-up, and small GEO immunotherapy datasets were analyzed separately as exploratory response-association cohorts. Spatial context was assessed using 10x Visium sections and Nanostring CosMx SMI data, with inference emphasized at the patient, section, or sample-contrast level rather than at the single-cell count level. Mendelian-randomization analyses were used for gene prioritization, not as definitive causal or therapeutic evidence.

Results: Nested cross-validation identified a stable 95-feature set enriched for immune and macrophage-context signals. C1QA, C1QB, and C1QC showed reproducible spatial autocorrelation in available spatial datasets (Moran's I approximately 0.35–0.39), whereas SPP1 showed stronger spatial clustering

(Moran's $I = 0.712$), consistent with a distinct macrophage-associated exclusion context. Retrospective model benchmarking showed statistically detectable but modest prognostic discrimination in TCGA and attenuated external performance in GSE31210; therefore, the model is reported as exploratory model behavior rather than as a validated clinical prediction tool. Small immunotherapy-treated cohorts showed directional response associations, but sample size and tumor-type heterogeneity preclude clinical claims.

Conclusions: C1Q/SPP1 captures a reproducible macrophage-spatial context in public NSCLC resources and may serve as a traceability feature for immune-active versus immune-excluded tumor-microenvironment states. The findings are hypothesis-generating and require prospective, spatially resolved, and experimental validation before clinical implementation or therapeutic targeting can be considered.

Keywords: Non-small cell lung cancer, Cancer informatics, C1Q, SPP1, macrophage context, spatial transcriptomics, reproducibility, traceability feature

1 Introduction

Non-small cell lung cancer (NSCLC) remains a major cause of cancer mortality, and immune checkpoint blockade has improved outcomes for a subset of patients [1–5]. However, PD-L1 expression, tumor mutational burden, and bulk immune-expression summaries do not fully capture the spatial and cellular organization of the tumor microenvironment (TME) [8, 9]. Spatial transcriptomics and single-cell technologies now make it possible to examine whether immune programs visible in bulk data also map to reproducible spatial contexts [14–16].

C1q, encoded by *C1QA*, *C1QB*, and *C1QC*, is produced largely by macrophage and dendritic-cell lineages in the TME. Prior work suggests that complement-associated macrophage states may accompany either immune activation or tumor-promoting inflammation, depending on context. *SPP1*-positive macrophage programs, by contrast, are often linked to stromal remodeling and immune exclusion. These observations motivate a narrower and more testable question: can the C1Q/SPP1 macrophage axis be traced reproducibly across public NSCLC transcriptomic, spatial, and clinical resources?

The present study is intentionally framed as a public cancer-informatics evidence synthesis. We do not claim to establish a clinical diagnostic assay, a treatment-selection tool, or a definitive mechanism. Instead, we evaluate whether C1Q/SPP1 behaves as a reproducible traceability feature linking macrophage context, spatial organization, retrospective survival associations, and exploratory immunotherapy-response signals. This conservative framing separates data-supported observations from hypotheses that require prospective and experimental validation.

2 Methods

2.1 Study Design and Evidence Hierarchy

This study used a staged, retrospective public-data design [25–27]. The evidence hierarchy was defined before interpretation: (1) TCGA NSCLC was used for discovery and internal model-behavior assessment; (2) GEO cohorts were used for external prognostic traceability where complete follow-up was available; (3) Visium and CosMx SMI datasets were used to evaluate spatial context, with inference emphasized at the patient, section, or paired-contrast level; and (4) immunotherapy-treated cohorts were treated as exploratory response-association datasets because of small sample sizes and tumor-type heterogeneity (Figure 1).

2.1.1 Data Acquisition and Quality Control

Discovery Cohort (TCGA): TCGA LUAD and LUSC transcriptomic and clinical data were retrieved from the NCI Genomic Data Commons [28, 29]. A total of 1,100 TCGA NSCLC tumor samples were initially considered (595 LUAD and 505 LUSC). After quality control and availability filtering, 1,038 samples contributed to discovery analyses. Analyses requiring complete survival, stage, or model features used the corresponding complete-case subsets; denominators are reported with each analysis.

Spatial Transcriptomics Data: For spatially resolved context analysis, we incorporated 10x Genomics Visium and Nanostring CosMx SMI datasets. Quality control, normalization, and platform-specific processing are described in the Supplementary Methods and source-data manifest.

- **10x Visium:** Included 8 tissue sections from 1 patient (Supplementary Table S12), utilizing a 55 μm spot diameter for tissue-level spatial mapping.
- **CosMx SMI:** Encompassed 8 tissue sections from 5 NSCLC patients, containing approximately 800,000 profiled cells targeted with a 960-gene RNA panel. Cell counts were treated as descriptive observations; independent inference was anchored to samples, sections, and paired contrasts.

External Cohorts: Clinical and expression data were retrieved from GEO. GSE31210 contained 226 lung adenocarcinoma samples after exclusion of non-tumor controls and 178 patients with complete follow-up for survival analysis. GSE91061, GSE126044, and GSE135222 were used only for exploratory immunotherapy-response association; GSE91061 was interpreted as cross-tumor support because it is not an NSCLC-only cohort.

2.2 C1Q/SPP1 Traceability Framework

We implemented a multi-stage traceability framework to evaluate whether C1Q/SPP1 macrophage context is reproducible across data layers:

Phase 1: MR-based gene prioritization: Mendelian randomization (MR) was used to prioritize genes with genetically supported associations [33], not to prove mechanism or treatment relevance. We used eQTL data from eQTLGen [35] and GTEx v8 with TwoSampleMR (v0.5.6), and inspected sensitivity diagnostics before assigning confidence.

Phase 2: Retrospective model-behavior assessment:	Machine-learning	185
	algorithms were benchmarked to assess whether the stable feature set carried survival-	186
	associated signal. The resulting models were not interpreted as clinical tools and no	187
	fixed clinical cutoff is proposed.	188
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Phase 3: Spatial context characterization:	A C1Q/SPP1-centered scoring	192
	approach was used to evaluate macrophage-associated spatial organization and CD8-	193
	marker context in Visium and CosMx data.	194
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2.3 Neighborhood Enrichment Analysis		198
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	Neighborhood enrichment analysis was performed in CosMx SMI data to describe local	200
	co-occurrence between C1Q-associated macrophage states and CD8-marker-positive	201
	contexts. A 50 μ m radius was used around each index cell, and observed neighbor-	202
	hood composition was compared with a permutation-based null distribution (N=1,000	203
	permutations). Because the number of independent patients was limited, these results	204
	were interpreted as local spatial-context support rather than definitive evidence of	205
	cell-cell signaling or causal recruitment.	206
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2.4 Data Preprocessing and Statistical Analysis		213
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2.4.1 Transcriptome Data Preprocessing		215
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	Bulk transcriptomic datasets were processed using dataset-appropriate normalization.	217
	TCGA RNA-seq expression values were filtered for low expression, log-transformed	218
	where appropriate, and aligned by gene symbol. External microarray data were	219
	quantile normalized or used as normalized series-matrix data when provided by	220
	GEO. Cross-platform analyses used within-cohort z-score scaling after feature align-	221
	ment; platform-specific effects were not interpreted biologically. Excluded samples and	222
	feature-retention decisions are documented in the source-data manifest.	223
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2.4.2 Statistical Analysis

All statistical analyses were performed using R and Python. Python-based data manipulation and visualization used Pandas [37], NumPy [38], SciPy [39], Matplotlib [40], and Seaborn [41]. Machine-learning tasks used Scikit-learn [42]. Survival analysis used Kaplan–Meier curves and Cox proportional hazards models. Model performance was evaluated with time-dependent AUC, concordance index (C-index), and hazard ratios with 95% confidence intervals where available. Data and code availability are detailed in the Declarations section.

2.5 Sample Size and Statistical Power

The TCGA discovery cohort (N=1,038 after quality control) was adequate for exploratory feature discovery and internal model-behavior assessment. External cohorts were smaller and had lower event counts; therefore, external survival and immunotherapy-response results were interpreted as traceability or exploratory association rather than confirmatory validation. No prospective sample-size calculation was possible because the study used public retrospective data.

2.6 Nested Cross-Validation Framework

To reduce data leakage and estimate feature stability, we implemented a 5×5 nested cross-validation framework.

Outer Loop (Model Evaluation):

- 5-fold stratified cross-validation (StratifiedKFold)
- Stratification basis: Vital status (alive/dead)
- Each fold: Training set 880 cases (80%), Test set 220 cases (20%)
- Random seed: 42 (ensuring reproducibility)

Inner Loop (Feature Selection):

• 5-fold cross-validation within each outer training set	277
• LASSO-based feature selection with α optimization using 5-fold cross-validation	278
and 10-fold grid search (α range: 10^{-5} to 10^{-1} , step size: 0.01). Optimal alpha was	279
selected based on minimum cross-validation error, with final $\lambda = 0.002$ yielding 91	280
non-zero coefficients (Supplementary Table S2).	281
• Features selected in 100% of outer folds (frequency = 5/5) retained as “core features”	282
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This double-layer strategy separated feature selection from held-out fold assess-	284
ment within TCGA. It does not replace independent prospective validation.	285
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2.7 Mendelian Randomization for Gene Prioritization	292
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We used two-sample Mendelian randomization [44] (MR), implemented in the	294
<i>TwoSampleMR</i> R package, as a gene-prioritization layer. MR estimates were inter-	295
preted as genetically supported associations under instrumental-variable assumptions,	296
not as proof of spatial mechanism, therapeutic actionability, or clinical utility.	297
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Instrumental Variable Selection	300
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• Genome-wide significance threshold: $P < 5 \times 10^{-8}$	302
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• Linkage disequilibrium (LD) pruning: $r^2 < 0.001$, window size = 10,000 kb	304
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• F-statistics calculated for all IVs, with $F > 10$ indicating strong instruments	306
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MR Methods	309
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• Inverse Variance Weighted (IVW): Primary summary estimate under standard	311
MR assumptions.	312
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• MR-Egger: Used to assess directional pleiotropy and provide a sensitivity estimate	314
under weaker assumptions.	315
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• Weighted Median: Used as a sensitivity estimate when a subset of instruments	317
may be invalid.	318
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Comprehensive Sensitivity Analyses:	321
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1. **Instrument strength:** F-statistics (median $F = 52,096.3$, 100% strong IVs with $F > 10$)
2. **Horizontal pleiotropy:** MR-Egger intercept test (18.5% of genes showed no pleiotropy, $P > 0.050$)
3. **Heterogeneity:** Cochran's Q test (62.3% showed no significant heterogeneity, $P > 0.050$)
4. **Outlier detection:** MR-PRESSO identified and removed outlier variants (70.0% of genes had no outliers, $P > 0.050$)
5. **Leave-one-out analysis:** Iteratively excluded each SNP to assess single-variant influence.

Exposure and Outcome Definitions: In all MR analyses, the exposure was defined as eQTL-mediated gene expression levels in immune-specific contexts, and the outcome was defined as the proportion of immune cell infiltration (e.g., CD8⁺ T-cell density) or specific immune phenotypes in lung cancer tissues.

Detailed MR diagnostic parameters and leave-one-out sensitivity plots are provided in Supplementary Table S11 and Supplementary Figures S9–S10. Candidates with pleiotropy, heterogeneity, weak instruments, or bidirectional signals were interpreted cautiously.

2.8 Spatial Transcriptomics Analysis

Recent studies have demonstrated the utility of deep learning in spatial transcriptomics [45, 46].

2.8.1 Spatial Autocorrelation Analysis

We quantified spatial clustering of gene expression using Moran's I statistic implemented in the 'spdep' R package, a measure of spatial autocorrelation ranging from -1 (dispersion) to +1 (clustering). We calculated Moran's I values for candidate genes

across spatial locations using a queen contiguity matrix with 10,000 permutations. Genes with Moran’s $I > 0.35$ and corrected $P < 0.001$ were described as spatially clustered. The term “hotspot” is used descriptively and does not imply causal organization.

2.8.2 Cell Type Deconvolution

Cell-type context in Visium data was estimated using the GigaTIME framework and cross-checked against canonical marker expression, including CD68 for macrophage context and CD8A/CD8B for CD8-marker context. These outputs were used as contextual estimates rather than direct ground-truth cell counts. CosMx SMI was used to inspect single-cell-resolution expression and neighborhood patterns in the available sections.

2.8.3 Spatial Co-localization Analysis

To assess spatial context among C1Q, SPP1, and immune-marker programs, we calculated correlations and local spatial statistics across multiple spatial scales ($50\mu\text{m}$, $100\mu\text{m}$, and $200\mu\text{m}$ where available) [47, 48]. Bivariate Moran’s I , Geary’s C , and local indicators of spatial association were used as descriptive spatial summaries with permutation-based significance testing and Benjamini–Hochberg correction. These analyses were interpreted as evidence of co-localization or spatial separation, not as proof of direct biological interaction.

2.9 Counterfactual Scoring and Sensitivity Analysis

We performed a counterfactual scoring analysis centered on *C1QA*. *C1QA* abundance was reduced computationally and the C1Q composite score, CD8 proxy, Moran’s I , network summaries, and risk-score outputs were recomputed. This procedure tests score dependency and internal coherence of the C1Q module; it is not an experimental perturbation and is not interpreted as intervention-like evidence. Dose-response,

mediation-style decomposition, bidirectional MR, Steiger directionality comparison, spatial scale-dependency, and specificity analyses were used as sensitivity analyses to assess whether the C1Q/SPP1 signal was internally consistent across analytic views.

2.10 Machine Learning Model Development and Evaluation

We systematically evaluated seven machine learning algorithms for retrospective survival modeling: AdaBoost, Random Forest, XGBoost, Logistic Regression, Support Vector Machine (SVM), Soft Voting ensemble, and LASSO Cox Regression. All models were trained on the TCGA cohort using the 95 core features identified through nested CV.

2.10.1 Model Training and Hyperparameter Optimization

Each algorithm underwent hyperparameter optimization using Bayesian optimization with 5-fold stratified cross-validation, implemented in the Optuna framework (version 3.1.0). The optimization objective was to improve balanced accuracy while controlling model complexity. We employed a Tree-structured Parzen Estimator (TPE) sampler for search-space exploration.

The hyperparameter tuning process was conducted as follows:

1. **Search Space Definition:** For each algorithm (AdaBoost, Random Forest, XGBoost, Logistic Regression, and SVM), we defined a search space covering key structural and regularization parameters.
2. **Optimization:** We performed 100 iterations of Bayesian optimization. To assess stability, this process was repeated 3 times with different random seeds.
3. **Model Selection:** The median-performing parameter set from the repeated runs was selected as the final configuration for downstream comparison.

Detailed search spaces and the final optimal hyperparameters for all models are listed in Supplementary Table S1.

2.10.2 External Evaluation	461
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To assess transferability beyond TCGA, we evaluated the feature set in GSE31210 as	463
an independent prognostic traceability cohort:	464
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• GSE31210 (N=178): Microarray gene expression data from primary lung adeno-	467
carcinomas with clinical follow-up	468
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Cross-platform Normalization Protocol: Raw data from different platforms	471
underwent harmonization using the following steps:	472
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1. Quantile normalization to standardize distribution across arrays/samples	475
2. Batch effect correction using ComBat-seq for RNA-seq data and sva for microarrays	476
3. Feature alignment by gene symbol matching with Ensembl ID resolution for	477
ambiguous symbols	478
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4. Z-score transformation of feature values within each dataset to account for	480
platform-specific scaling	481
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Evaluation Metrics: Model behavior was evaluated using the same retrospective	485
metrics as the training cohort where endpoint availability allowed:	486
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• Discrimination: Time-dependent AUC at 1, 3, and 5 years	489
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• Calibration: Calibration plots with Hosmer-Lemeshow test	491
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• Risk stratification: Hazard ratios and Kaplan-Meier survival curves	493
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• Cross-cohort consistency: direction and approximate magnitude of the risk-score	495
association	496
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• Heterogeneity checks: interaction terms between cohort and risk score in combined	498
Cox models where harmonized covariates were available	499
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Transferability was judged descriptively from discrimination, survival separa-	501
tion, and directionally compatible hazard ratios. These criteria were pragmatic and	502
exploratory, not pre-registered clinical benchmarks.	503
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2.10.3 Performance Evaluation

Model performance was assessed using multiple retrospective metrics:

- **Discrimination:** Area Under the ROC Curve (AUC), Accuracy, Precision, Recall, F1-score
- **Ranking:** Concordance index (C-index)
- **Decision support:** Decision curve analysis (DCA) for net benefit exploration
- **Survival analysis:** Hazard ratios (HR) with 95% confidence intervals, log-rank test P-values

Time-dependent ROC curves were generated for 1-year, 3-year, and 5-year survival predictions. Risk scores were calculated for each patient, and the cohort was stratified into high-risk and low-risk groups using the median risk score as a descriptive cutoff.

2.10.4 Model Selection Rationale

While XGBoost achieved the highest AUC (0.781) in internal evaluation, we retained AdaBoost as the reference model for downstream description because it produced stable survival separation and a compact risk formulation. This is a pragmatic modeling choice, not evidence that AdaBoost is uniformly superior across all metrics. The ensemble nature of AdaBoost can be helpful for high-dimensional data, but no clinical superiority is claimed.

2.11 Statistical Analysis

Continuous variables were compared via Student's t-test or Mann-Whitney U test depending on normality (assessed by Shapiro-Wilk test). Categorical variables were compared using Chi-square test or Fisher's exact test. Survival analyses were performed using Cox proportional hazards regression, with both univariate and multivariate models adjusted for age, gender, and stage.

Correlation analyses employed Pearson correlation for normally distributed variables and Spearman correlation for non-normal distributions. Multiple testing correction was applied as follows:

- **Differential gene expression:** Benjamini-Hochberg FDR < 0.05
- **Mendelian randomization:** FDR < 0.05 (198 genes tested)
- **Spatial autocorrelation (Moran's I):** Bonferroni correction, $\alpha = 0.05/100 = 0.001$
- **Gene-immune marker correlations:** Benjamini-Hochberg FDR < 0.05
- **Pathway enrichment (GO/KEGG):** Benjamini-Hochberg FDR < 0.05

All statistical analyses were performed in R version 4.2.0 and Python 3.9.0. Two-sided P -values < 0.05 were considered statistically significant unless otherwise specified.

2.12 Hazard Ratio Calculation

Hazard ratios (HRs) and 95% confidence intervals (CIs) were calculated using univariate Cox proportional hazards regression implemented in the *lifelines* Python package (version 0.27.0). Risk scores from each machine learning model were obtained through 5-fold cross-validation, standardized using z-score transformation, and treated as continuous variables. HRs represent the hazard increase per 1 standard deviation increase in risk score. Wald 95% confidence intervals were computed, and P -values were obtained from the likelihood ratio test. For visualization purposes in Kaplan-Meier curves, patients were stratified into high-risk and low-risk groups based on the median risk score, and survival differences were assessed using the log-rank test.

2.13 Prognostic Model Construction and Survival Analysis

To evaluate whether the stability-prioritized feature set carried retrospective outcome signal, we constructed a descriptive prognostic score using AdaBoost with optimized hyperparameters (`n_estimators=100`, `learning_rate=0.1`, `random_state=42`). AdaBoost

599 was used as a reference model for downstream description rather than as a clinically
600 preferred algorithm. Patients were stratified into high- and low-score groups using the
601 median risk score for visualization only. Kaplan–Meier curves and log-rank tests were
602 used to display survival differences, and continuous standardized scores were used in
603 Cox models. Predictive accuracy was summarized with C-index and time-dependent
604 AUC.

609 610 **3 Results**

613 The results are organized according to the evidence hierarchy: cohort accounting,
614 feature stability, MR-based prioritization, spatial context, counterfactual sensitivity,
615 retrospective model behavior, external traceability, and exploratory immunotherapy-
616 response association.

619 620 **3.1 Study Population and Baseline Characteristics**

623 We initially considered 1,100 TCGA NSCLC tumor samples (595 LUAD, 505 LUSC).
624 After quality control and availability filtering, 1,038 samples contributed to discovery
625 and feature-screening analyses. The GSE31210 cohort included 226 tumor samples,
626 of which 178 had complete follow-up for survival analysis. Key baseline differences
627 included:

- 632 • **Gender disparity:** LUSC had higher male representation (74.1% vs. 41.3% in
633 LUAD)
- 635 • **Survival outcomes:** LUSC patients showed higher mortality (43.6% vs. 32.1% in
636 LUAD)

639 Detailed characteristics are provided in Table 1 and Supplementary Table S8.

Table 1 Baseline characteristics of study population (TCGA NSCLC cohort and GSE31210 traceability cohort)

Characteristic	Total (N=1100)	LUAD (N=595)	LUSC (N=505)	GSE31210 (N=226 ^b)
Age (years)	66.2 ± 9.4	65.1 ± 10.1	67.3 ± 8.6	59.6 ± 7.4
Gender				
Male	620 (56.4%)	246 (41.3%)	374 (74.1%)	105 (46.5%)
Female	417 (37.9%)	286 (48.1%)	131 (25.9%)	121 (53.5%)
Unknown	63 (5.7%)	63 (10.6%)	0 (0.0%)	0 (0.0%)
AJCC Stage				
Stage I	535 (48.6%)	289 (48.6%)	246 (48.7%)	168 (74.3%)
Stage II	288 (26.2%)	125 (21.0%)	163 (32.3%)	58 (25.7%)
Stage III	169 (15.4%)	84 (14.1%)	85 (16.8%)	0 (0.0%)
Stage IV	33 (3.0%)	26 (4.4%)	7 (1.4%)	0 (0.0%)
Unknown	75 (6.8%)	71 (11.9%)	4 (0.8%)	0 (0.0%)
Smoking History				
Light (<10 pack-years)	35 (3.2%)	22 (3.7%)	13 (2.6%)	N.A.
Moderate (10–30 pack-years)	170 (15.5%)	105 (17.6%)	65 (12.9%)	N.A.
Heavy (≥30 pack-years)	588 (53.5%)	238 (40.0%)	350 (69.3%)	N.A.
Unknown	307 (27.9%)	230 (38.7%)	77 (15.2%)	N.A.
Vital Status				
Alive	626 (56.9%)	341 (57.3%)	285 (56.4%)	191 (84.5%)
Dead	411 (37.4%)	191 (32.1%)	220 (43.6%)	35 (15.5%)
Unknown	63 (5.7%)	63 (10.6%)	0 (0.0%)	0 (0.0%)

Note: ^bThe GSE31210 cohort includes 226 primary lung adenocarcinoma samples after excluding normal tissue controls; survival analyses used 178 patients with complete follow-up data. N.A. = Not Available. ^a*P*-values compare TCGA (Total) vs. GSE31210 using Mann-Whitney U test for continuous variables and Chi-square test for categorical variables. The discovery cohort for feature screening consisted of 1,038 patients after QC.

3.2 Robust Feature Selection via Nested Cross-Validation

Nested cross-validation identified 95 stable features selected in 100% of outer folds, with 212 features showing >60% selection frequency (Figure 2C, Figure S2, Supplementary Table S4). These features were used as a stability-prioritized set for downstream exploratory modeling.

Key stable features comprised:

- **ACP5** (macrophage regulation);

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Table 2 Performance comparison of machine learning algorithms for prognostic model construction

Model	Test AUC	Test Accuracy	Test F1-score	C-index	HR (95% CI)	P-value
XGBoost	0.781	0.725	0.715	0.591	1.56 (1.33–1.84)	<0.001
Naive Bayes	0.680	0.679	0.680	0.680	1.59 (1.33–1.91)	<0.001
Soft Voting	0.757	0.738	0.732	0.604	1.59 (1.35–1.88)	<0.001
Random Forest	0.701	0.657	0.486	0.657	1.59 (1.34–1.89)	<0.001
SVM	0.727	0.667	0.493	0.656	1.51 (1.28–1.79)	<0.001
Logistic Regression	0.741	0.705	0.652	0.650	1.45 (1.23–1.71)	<0.001
Gradient Boosting	0.742	0.698	0.685	0.650	1.56 (1.33–1.84)	<0.001
AdaBoost	0.743	0.657	0.526	0.607	1.43 (1.21–1.70)	<0.001
K-Nearest Neighbors	0.689	0.654	0.645	0.620	1.37 (1.15–1.64)	<0.001
Decision Tree	0.612	0.598	0.601	0.564	1.29 (1.10–1.51)	0.002
LASSO Cox	0.725	0.686	0.629	0.558	1.52 (1.39–1.66)	<0.001

Note: Models were evaluated using 5-fold nested cross-validation. Test AUC represents area under the ROC curve on the held-out fold. C-index measures risk ranking. HR and 95% CI were calculated using continuous Cox regression with standardized risk scores. Bold values identify the reference AdaBoost model used for downstream descriptive analyses; they do not indicate uniform superiority across all metrics.

• <i>KCNAB2</i> (ion channel function); and	737
• <i>BOK</i> (pro-apoptotic pathway).	738
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Feature selection stability analysis confirmed clear separation between highly stable	740
(frequency = 5/5) and less consistent features (Figure S2A).	741
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3.3 Mendelian Randomization Prioritizes a Reproducible	745
Gene Set in NSCLC	746
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Mendelian randomization prioritized 198 genes as genetically supported candi-	748
dates associated with lung cancer immune characteristics (Figure 3, Figure S3, Figure	749
S9, Supplementary Tables S7 and S11). The diagnostics suggested that some candi-	750
dates had usable instruments, while others should be treated cautiously because of	751
heterogeneity, pleiotropy, or bidirectional signals. These results are better viewed as a	752
prioritization layer than as definitive causal proof.	753
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Key Findings	758
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• Hub Genes: <i>KCNAB2</i> and <i>C1QA</i> emerged as central network-linked candidates	760
in the MR-filtered set	761
• Genetic Associations: <i>HIST1H4F</i> (IVW $\beta = 0.251$, $P < 1 \times 10^{-200}$), <i>BOK</i> (IVW	762
$\beta = -0.119$, $P = 5.580 \times 10^{-9}$), <i>KCNAB2</i> (IVW $\beta = -0.078$, $P = 1.070 \times 10^{-13}$),	763
and <i>ACP5</i> (IVW $\beta = 0.234$, $P = 4.760 \times 10^{-9}$) were among the better-supported	764
associations	765
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• Robustness: Leave-one-out analysis suggested that no single SNP dominated the	770
top signals	771
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• Functional Enrichment: Significant enrichment in immune response pathways	775
(GO:0006955), T cell receptor signaling (hsa04660), and cytokine-cytokine receptor	776
interaction (hsa04060)	777
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3.4 Identification of Spatial C1Q Immune Hotspots

High-resolution spatial mapping reveals reproducible C1Q clustering. Utilizing both 10x Visium and CosMx SMI platforms, we observed that C1Q expression was spatially clustered rather than uniformly distributed within the available TME sections (Figures 4–5, Supplementary Figure S5, Table 3, Supplementary Table S5). C1Q family members exhibited moderate spatial autocorrelation among immune genes:

- **C1QA**: Moran's $I = 0.386$ ($P < 0.001$)
- **C1QC**: Moran's $I = 0.377$ ($P < 0.001$)
- **C1QB**: Moran's $I = 0.355$ ($P < 0.001$)

Global analysis of 100 immune-related genes placed C1Q family members among the more spatially clustered immune genes, alongside UBXLN10 (Moran's $I = 0.416$). SPP1 showed stronger spatial autocorrelation (Moran's $I = 0.712$, $P < 0.001$), consistent with a more spatially compartmentalized macrophage context.

Neighborhood enrichment analysis suggested that C1Q-associated macrophage contexts were often near CD8-marker-positive contexts. GigaTIME deconvolution linked C1Q regions to myeloid and T-cell marker enrichment, including CD14, CD163, CD68, and CD8-marker proxies ($P < 1 \times 10^{-6}$; Supplementary Table S6, Figure S6). CD138-positive plasma-cell context was analyzed separately and is not treated as a macrophage marker. These results support a spatial association, but not direct cell-cell signaling.

CosMx SMI context analysis across 8 sections from 5 patients showed broadly similar spatial patterns, although the cross-platform statistic was small in magnitude (bivariate Moran's $I = 0.009$). This supports limited reproducibility of the pattern rather than strong effect size.

3.5 Counterfactual Scoring Shows Dependency of the C1Q	829
Module	830
To test whether the C1Q score was internally dependent on <i>C1QA</i> , we performed a	831
counterfactual scoring analysis and re-estimated spatial and clinical metrics (Supple-	832
mentary Figures S11–S12). The computational attenuation reduced the C1Q program	833
score (mean C1Q score: 5.179 $\rightarrow$ 2.628) and decreased Moran’s I (0.660 $\rightarrow$ 0.401; Δ	834
= -0.259). This indicates score dependency and module sensitivity, not experimental	835
proof of hotspot collapse.	836
Additional sensitivity analyses showed a coherent pattern across several views.	837
Dose-response modeling demonstrated monotonic trends across C1Q quartiles, with	838
increasing CD8 proxy signal (trend slope = 0.478, $P = 6.27 \times 10^{-30}$) and coordi-	839
nated risk-score shift (trend slope = 0.586, $P = 4.35 \times 10^{-110}$). Bidirectional MR	840
identified forward and reverse signals for some candidates, which argues for caution	841
rather than overclaiming. Spatial scale analysis indicated the strongest local associa-	842
tion at the shortest tested radius. Mediation-style decomposition suggested a possible	843
indirect component through the CD8 proxy, while specificity testing ranked the	844
C1Q axis among the better-supported immune modules relative to non-C1Q controls	845
(Supplementary Figure S12).	846
The associated visualization and network-summary sensitivity outputs are pro-	847
vided in Supplementary Figures S11–S12 rather than treated as additional main-result	848
figures, because these analyses are sensitivity checks rather than experimental	849
perturbations.	850
3.6 Machine Learning Model Performance and Risk	851
Stratification	852
The C1Q/SPP1 feature set showed retrospective survival-associated model	853
behavior. Evaluation of seven machine learning algorithms showed that different	854

875 algorithms optimized different metrics (Table 2, Figure 2A–B, Supplementary Table
876 S1). AdaBoost was retained as the reference model for downstream description, but
877
878 it was not uniformly superior across all metrics.
879

880 **Key Performance Metrics**

- 881
- 882 • **AdaBoost** showed moderate ranking performance (C-index = 0.607) and a signif-
883 icant association between standardized risk score and survival (HR = 1.430, 95%
884 CI: 1.21–1.70, $P < 0.001$).
885
 - 886 • **XGBoost** achieved the highest AUC (0.781), whereas other algorithms showed
887 higher C-index values. This metric discordance argues against presenting any single
888 model as clinically superior.
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890
891

892 **Risk Stratification:** AdaBoost stratified a TCGA complete-case subset into high-
893 score and low-score groups with different survival outcomes (log-rank $P = 0.005$, HR
894 = 1.620, 95% CI: 1.15–2.27). To avoid ambiguity, higher model risk score denotes the
895 adverse-risk direction; C1Q-enriched immune-active context is interpreted separately
896 from the aggregate risk-score direction. Time-dependent ROC showed modest discrim-
897 ination: 1-year AUC of 0.609, 3-year AUC of 0.611, and 5-year AUC of 0.590 (Figure
898 6A–B).
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901 **External Traceability:** In GSE31210, the feature set showed attenuated but
902 directionally compatible performance (AUC = 0.650, C-index = 0.629; Figure 7, Sup-
903 plementary Figure S7). Survival separation was sensitive to analysis specification,
904 with source tables reporting both significant and borderline estimates depending on
905 completeness and covariate handling. Therefore, GSE31210 is treated as external trace-
906 ability rather than definitive validation. Decision curve analysis is reported only as
907 exploratory net-benefit visualization.
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3.7 Exploratory Immunotherapy-Response Association

We next examined whether the C1Q/SPP1-associated score tracked response in small immunotherapy-treated datasets. These analyses are exploratory because cohort sizes were limited and GSE91061 is not NSCLC-specific. We therefore report directional associations without claiming clinical response prediction or superiority over PD-L1/TMB.

GSE91061 (melanoma-dominant exploratory cohort): Low-risk patients showed higher objective response rates (42.3% vs. 15.8%, $P = 0.018$) with AUC = 0.758. Responders exhibited decreasing risk scores during therapy (mean change: -0.210 ± 0.150 , $P = 0.006$). Because this cohort is not NSCLC-specific, it is treated as cross-tumor exploratory support.

GSE126044 (NSCLC): The risk score discriminated responders with AUC = 0.813 and was associated with longer PFS in low-risk patients (median 12.4 vs. 3.2 months, HR = 0.280, $P = 0.018$), but the cohort size was small (N=16).

GSE135222 (NSCLC): A directional response association was observed with AUC = 0.806 ($P = 0.003$), again limited by small sample size (N=27).

Subtype summaries suggested that lower-risk or C1Q-enriched contexts were more often aligned with inflamed immune programs, whereas higher-risk contexts were more often aligned with excluded or desert-like features, including SPP1-associated macrophage programs. These subtype assignments are descriptive and require independent validation.

3.8 Multivariable Survival Association

To assess whether the score retained survival association after adjustment for available clinical variables, we performed Cox regression analyses (Table 4, Supplementary Table S9).

Univariate Analysis

- **Risk Score:** HR = 2.450 (95% CI: 1.85–3.24), $P < 0.001$
- **Age** (per year): HR = 1.020 (95% CI: 1.00–1.03), $P = 0.064$ (marginally significant)
- **Gender** (Male vs. Female): HR = 0.980 (95% CI: 0.73–1.32), $P = 0.898$ (not significant)
- **Stage** (Stage IV vs. Stage I): HR = 1.330 (95% CI: 1.09–1.61), $P = 0.005$ (significant)

Multivariate Analysis

- **Risk Score:** HR = 1.430 (95% CI: 1.21–1.70), $P < 0.001$
- **Age** (per year): HR = 1.010 (95% CI: 0.99–1.03), $P = 0.217$ (not significant)
- **Gender** (Male vs. Female): HR = 0.900 (95% CI: 0.67–1.21), $P = 0.482$ (not significant)
- **Stage** (Stage IV vs. Stage I): HR = 1.350 (95% CI: 1.11–1.65), $P = 0.003$ (significant)

Multivariate Analysis (Supplementary Table S9): After adjusting for age, gender, and stage, the risk score retained association with survival endpoints in the available complete-case models:

- **Overall Survival:** Age HR = 1.020 (95% CI: 0.98–1.06), Gender HR = 1.150 (95% CI: 0.72–1.84), Stage II vs. Stage I HR = 2.340 (95% CI: 1.45–3.78)
- **Recurrence-Free Survival:** Age HR = 1.010 (95% CI: 0.97–1.05), Gender HR = 1.080 (95% CI: 0.68–1.72), Stage II vs. Stage I HR = 2.870 (95% CI: 1.78–4.62)

3.9 Integration of MR-Prioritized Genes and Spatial

Correlations

Integration of MR-prioritized gene sets with spatial transcriptomics identified candidates with both genetic-association and spatial-context support.

KCNAB2 emerged as a recurrent candidate with 100% selection frequency in nested CV and measurable, though modest, spatial clustering.

Additional Integrated Candidates

- **LINC00339** (long non-coding RNA): Network degree of 0.064, correlation $r = 0.586$ ($P < 0.001$)
- **CDC42** (signal transduction): Network degree of 0.060, correlation $r = 0.424$ ($P < 0.001$)
- **PLOD1** (collagen modification): Network degree of 0.035, correlation $r = 0.260$ ($P < 0.001$)
- **UBXN10** (ubiquitination): Highest Moran's I = 0.416 ($P < 0.001$)

Table 3 Spatial and MR-prioritized characteristics of candidate genes

Gene	Annotation	Selection Stability	Moran's I	Network Degree	Correlation
ACP5	Macrophage Reg.	100% (5/5)	N.D.	-	-
KCNAB2	Ion Channel	100% (5/5)	0.094***	0.078	0.219***
SPP1	TME Modulation	-	0.712***	0.066	-0.258***
LINC00339	Long ncRNA	-	N.D.	0.064	0.586***
CDC42	Signal Transduction	-	N.D.	0.060	0.424***
PLOD1	Collagen Modification	-	N.D.	0.035	0.260***
C1QA	Complement	-	0.386***	-	-
C1QB	Complement	-	0.353***	-	-
C1QC	Complement	-	0.377***	-	-
UBXN10	Ubiquitination	-	0.416***	-	-
BOK	Apoptosis	100% (5/5)	N.D.	-	-

Note: N.D. = Not Determined; *** $P < 0.001$. Network degree centrality derives from the MR-prioritized network. Correlation represents Pearson correlation coefficient with immune-infiltration estimates.

3.10 Public Protein-Expression Context

We cross-referenced public immunohistochemistry images for protein-level context.

KCNAB2 showed cytoplasmic staining in available lung tissue images, while SPP1

Table 4 Univariate and multivariable Cox regression analysis in the TCGA discovery cohort (N=1,038)

Covariate	Univariate Analysis		Multivariate Analysis	
	HR (95% CI)	P-value	HR (95% CI)	P-value
Risk Score ^a	2.45 (1.85–3.24)	< 0.001	1.43 (1.21–1.70)	< 0.001
Age	1.02 (1.00–1.03)	0.064	1.01 (0.99–1.03)	0.217
Gender (Male vs Female)	0.98 (0.73–1.32)	0.898	0.90 (0.67–1.21)	0.482
Stage (IV vs I)	1.33 (1.09–1.61)	0.005	1.35 (1.11–1.65)	0.003

Note: Univariate and multivariable Cox regression analysis in the TCGA cohort including risk score, age, gender, and AJCC stage. ^aRisk score was standardized before analysis. The HR represents the change in risk per standard-deviation increase in risk score. These retrospective associations require external and prospective validation before clinical interpretation. For GSE31210 traceability results, see Supplementary Table S9.

was predominantly stromal in the inspected examples (Figure 8). These data support plausibility of protein-level expression patterns but should not be interpreted as independent experimental validation.

4 Discussion

4.1 Principal Findings

This study supports a conservative conclusion: C1Q/SPP1 is a reproducible macrophage-context traceability feature in public NSCLC resources. Across bulk transcriptomics, MR-based prioritization, spatial datasets, and retrospective outcome cohorts, the C1Q/SPP1 axis repeatedly marked immune-active versus macrophage-compartmentalized states. The evidence is internally coherent, but it does not establish a clinical assay, a treatment-selection rule, or a definitive mechanism.

The principal findings were: (1) nested cross-validation identified a stable 95-feature set enriched for immune and macrophage-context signals; (2) MR prioritized several candidates, including *C1QA* and *KCNAB2*, but with the usual instrumental-variable caveats; (3) C1Q-family genes showed moderate spatial clustering, while *SPP1* showed stronger spatial compartmentalization; (4) counterfactual scoring showed that

the C1Q module is internally sensitive to *C1QA* attenuation, without proving biological perturbation; and (5) retrospective model behavior was statistically detectable but modest, with external and immunotherapy-response results best interpreted as traceability rather than validation.

4.2 Interpretation of the Evidence Chain

The strength of the study is not a single high-performing classifier. Rather, it is the convergence of several independent public-data views on the same macrophage-context axis. Bulk expression suggested a survival-associated immune program; MR prioritized genes that intersected with this program; spatial data placed C1Q and SPP1 in non-random local contexts; and small immunotherapy-treated cohorts showed directional response associations. This layered consistency makes the C1Q/SPP1 axis biologically plausible and worth further study.

At the same time, each layer has limits. MR is not mechanism. Spatial autocorrelation is not cellular communication. A computational C1QA attenuation experiment is not a wet-lab perturbation. Small ICB cohorts are not clinical validation. For that reason, the manuscript should be read as a reproducible hypothesis-generating cancer-informatics study rather than as a completed biomarker-development paper.

4.3 Biological Plausibility of the C1Q/SPP1 Spatial Context

The biological plausibility of the C1Q/SPP1 axis rests on macrophage biology and spatial organization. Recent spatial transcriptomics studies have demonstrated the importance of spatial context in understanding immune microenvironments. Our finding that C1Q-associated regions track immune-marker context suggests that complement-associated macrophage states may mark immune-active local environments. This is a marker-context interpretation, not proof that C1Q itself drives therapeutic response.

1151 The complement system, traditionally viewed as part of innate immunity against
1152 pathogens, has emerged as a critical regulator of tumor immunity . B cells have also
1153 been associated with survival and immunotherapy response in other cancers [49, 50].
1154 Studies have shown that complement components play dual roles in tumor immunity
1155 . C1Q can influence tumor progression through multiple mechanisms: (1) modula-
1156 tion of T cell responses, (2) regulation of macrophage polarization, (3) promotion of
1157 angiogenesis, and (4) facilitation of tumor cell migration and metastasis.
1158

1159 In our study, C1Q-associated macrophage contexts were often located near
1160 CD8-marker-positive contexts. This pattern is compatible with immune-active local
1161 environments, but it cannot distinguish recruitment, retention, shared upstream
1162 inflammation, or sampling structure. The appropriate conclusion is that C1Q marks
1163 a spatial immune context that deserves mechanistic follow-up.
1164

1165 The stronger spatial autocorrelation of *SPP1* (Moran's $I = 0.712$, $P < 0.001$)
1166 delineated a distinct macrophage-associated compartment. This is consistent with
1167 prior descriptions of *SPP1*-positive macrophage programs in stromal remodeling and
1168 immune exclusion, but our data do not prove that *SPP1* creates a physical bar-
1169 rier in NSCLC. The C1Q/*SPP1* contrast should therefore be framed as a candidate
1170 immune-active versus macrophage-compartmentalized context.

1171 The spatial correlation between C1Q family members and macrophage markers
1172 indicates that these clusters are macrophage-contextual rather than random expression
1173 artifacts. Beyond C1Q and *KCNAB2*, the integrated framework identified additional
1174 candidates:
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1176 • ***SPP1* (osteopontin)** exhibited the strongest spatial clustering (Moran's $I =$
1177 0.712) among the candidates and is a plausible marker of macrophage-associated
1178 stromal or exclusion context.
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• <i>BOK</i> , a BCL-2 family member, showed 100% selection frequency in nested CV	1197
and MR support (IVW $\beta = -0.119$, $P = 5.58 \times 10^{-9}$), making it a candidate for	1198
follow-up rather than an established driver.	1199
	1200
• <i>HIST1H4F</i> , a histone H4 variant, may reflect epigenetic or proliferative context.	1201
Its role in the C1Q/SPP1 axis remains speculative without direct functional assays.	1202
	1203
Together, these patterns argue that bulk expression signatures should be inter-	1204
preted with spatial and cellular context. They do not yet justify a diagnostic or	1205
therapeutic strategy.	1206
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4.4 KCNAB2 as a Candidate Context Feature	1212
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<i>KCNAB2</i> , a potassium channel subunit, emerged as a recurrent candidate in feature	1214
selection and MR-prioritized analyses. Its appearance across multiple analytic layers	1215
suggests that ion-channel-related biology may intersect with immune context, but the	1216
present data do not establish <i>KCNAB2</i> as a driver of TME function.	1217
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The co-occurrence of <i>KCNAB2</i> with C1Q-associated immune contexts (Figure 8C)	1220
raises a testable hypothesis that potassium-channel-related programs may track	1221
macrophage or immune-effector state. This remains a hypothesis. The current analy-	1222
sis cannot determine whether <i>KCNAB2</i> acts in macrophages, T cells, tumor cells, or	1223
another compartment.	1224
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	1228
The MR-prioritized association for <i>KCNAB2</i> is biologically interesting, but it	1229
should not be treated as causal proof. Figure 8C provides context for expression	1230
distribution, while definitive cell-type localization and functional direction require	1231
orthogonal validation.	1232
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We therefore describe <i>KCNAB2</i> as a candidate context feature rather than a	1236
metabolic checkpoint. Future studies could test whether <i>KCNAB2</i> expression reflects	1237
immune-cell state, tumor-cell state, stromal composition, or technical correlation with	1238
broader immune infiltration.	1239
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1243 Based on our findings, the most defensible hypothesis is that C1Q-enriched
1244 macrophage contexts, *SPP1*-enriched macrophage contexts, and *KCNAB2*-linked
1245 programs may represent partially distinct immune microenvironment states.
1246

1247 These observations justify follow-up but not therapeutic targeting. The candi-
1248 date status of *KCNAB2* should be tested using cell-type-resolved protein assays,
1249 perturbation experiments, and prospective outcome cohorts.
1250

1251 Our decision to highlight *KCNAB2* was driven by its selection stability and
1252 MR-prioritized association. This makes it a candidate for further validation, not an
1253 established therapeutic target.
1254

1255 Based on our findings, we propose two key hypotheses for the role of *KCNAB2* in
1256 the NSCLC TME:
1257

- 1258 1. **Metabolic-Immune Coupling:** *KCNAB2* may be related to the metabolic fitness
1259 of macrophages in C1Q-enriched contexts by modulating potassium efflux, thereby
1260 supporting a plausible follow-up hypothesis in nutrient-depleted TME states.
1261
- 1262 2. **Spatial Scaffolding:** The spatial co-localization of *KCNAB2* and C1Q suggests
1263 a structural role for ion channel regulators in organizing immune clusters, possibly
1264 through the regulation of cell adhesion or cytokine secretion.
1265

1266 4.5 Machine Learning Model Behavior and Its Limits

1267 Our systematic evaluation of seven machine-learning algorithms showed that model
1268 performance depends strongly on the metric used. XGBoost achieved the highest
1269 AUC (0.781), whereas several models had higher C-index values than AdaBoost. For
1270 this reason, the model comparison should not be used to claim clinical superiority
1271 of one algorithm. AdaBoost is retained only as a reference model for describing one
1272 reproducible score behavior.
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The modest time-dependent AUC values and moderate C-index indicate that the model captures some prognostic signal but is not ready for individual patient decision-making. The score is useful here as a traceability device for testing whether the C1Q/SPP1-associated feature set carries outcome-linked information.

Decision curve analysis is included as an exploratory visualization of possible net benefit. It should not be interpreted as evidence that the model can guide treatment decisions.

The external evaluation in GSE31210 was directionally supportive but not definitive. Depending on the analysis specification, the survival estimate was significant or borderline, with wide confidence intervals. This pattern is compatible with a weak-to-moderate transferable signal, but it is not enough for clinical validation.

Translation Boundary: The current evidence does not support clinical implementation. A future clinical-grade model would require a locked feature set, a locked cutoff, prospective validation, calibration assessment, decision-curve confirmation in clinically relevant populations, and comparison with stage-only and clinical-only baselines.

Biological Interpretation: Nonlinear models may capture interactions among immune-context features, but algorithmic performance cannot be used as evidence of mechanism. The model results support a reproducible association between macrophage-context features and outcome-linked states.

Prognostic Association: The model demonstrates statistically detectable prognostic association in retrospective data (C-index 0.607, HR 1.43, 95% CI: 1.21–1.70, $P < 0.001$), but its incremental value beyond routine clinical factors remains to be established rigorously.

Immunotherapy Response Association: The immunotherapy-treated cohorts show directional associations (AUC range: 0.758–0.813), but their small size and

tumor-type heterogeneity preclude claims of robust ICI prediction or superiority over PD-L1/TMB.

The longitudinal pattern in GSE91061 is compatible with dynamic immune remodeling during treatment, but because the cohort is not NSCLC-specific, this observation should be treated as exploratory.

The integration of spatial features with MR-prioritized candidates addresses a limitation of bulk-only interpretation, but the present implementation remains retrospective and hypothesis-generating.

4.6 Future Translation Requirements

For this work to move toward clinical use, several requirements must be met.

Future studies should prioritize:

1. locking the feature set, preprocessing pipeline, model, and cutoff before testing;
2. validating the model in NSCLC-only immunotherapy cohorts with adequate sample size and event counts;
3. comparing against clinical-only, stage-only, PD-L1, TMB, and published immune-signature baselines;
4. confirming C1Q/SPP1 protein-level spatial context using multiplex FFPE-compatible assays;
5. experimentally testing whether C1Q/SPP1-associated macrophage states have functional effects on T-cell localization or activity.

Only after these steps would a simplified pathological or molecular panel be justified.

4.6.1 Biological Rationale for Resistance	1381
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Our TME subtype analysis suggested that higher-risk contexts were enriched for	1383
immune-exclusion-associated programs, including SPP1-associated macrophage fea-	1384
tures. These observations may help generate hypotheses about ICI resistance, but they	1385
do not explain resistance mechanisms on their own.	1386
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4.6.2 Limitations and Future Directions	1391
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Several limitations require emphasis. The primary survival analyses are retrospective	1393
and rely heavily on overall survival, which is affected by treatment heterogeneity,	1394
comorbidity, stage, and incomplete clinical annotation. The spatial datasets include	1395
few independent patients, and the large CosMx cell count should not be mistaken	1396
for a large number of independent biological replicates. The Visium dataset provides	1397
spatial context but not single-cell precision.	1398
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Regarding immunotherapy-response association, several considerations warrant	1401
discussion:	1402
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1. The GSE91061 cohort includes melanoma and is not NSCLC-specific. It provides	1407
cross-tumor exploratory support only.	1408
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2. Spatial transcriptomics analysis was performed largely on treatment-naïve sam-	1410
ples. On-treatment biopsies would be required to assess spatial remodeling during	1411
therapy.	1412
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3. The small sample sizes of immunotherapy cohorts (GSE126044: N=16; GSE135222:	1415
N=27) limit precision and subgroup analysis. Prospective validation is essential	1416
before clinical implementation.	1417
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4. Integration with PD-L1, TMB, ctDNA dynamics, TCR repertoire, and treatment	1420
metadata will be needed before incremental value can be assessed.	1421
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1427 Future translational studies should prioritize a locked, FFPE-compatible assay;
1428 adequately powered NSCLC-only cohorts; prespecified cutoffs; and direct comparison
1429 with existing clinical and molecular biomarkers.
1430

1431 Additional limitations include population imbalance in TCGA, incomplete treat-
1432 ment data, cross-platform technical variance, possible residual confounding, and the
1433 absence of wet-lab perturbation. The counterfactual scoring analysis is a sensitivity
1434 analysis only. The model has not been validated in neoadjuvant immunotherapy or
1435 prospective randomized settings.
1436

1437 Overall, the study provides a reproducible public-data framework for tracing
1438 macrophage-context signals across molecular and spatial layers. Its most immediate
1439 value is hypothesis prioritization for future spatial and functional validation.
1440

1441 5 Conclusions

1442 This study identifies C1Q/SPP1 as a reproducible macrophage-context traceability
1443 feature in public NSCLC resources. C1Q-family genes showed moderate spatial clus-
1444 tering, SPP1 showed stronger spatial compartmentalization, and retrospective model
1445 behavior suggested outcome-linked signal with modest discrimination. MR and coun-
1446 terfactual analyses support prioritization but do not prove mechanism. The findings
1447 are hypothesis-generating and require prospective, NSCLC-specific, spatially resolved,
1448 and experimental validation before clinical implementation or therapeutic targeting
1449 can be considered.
1450

1451 Supplementary Information

1452 **Additional file 1: Supplementary Material.** Contains Supplementary Figures
1453 S1–S12 and Supplementary Tables S1–S13. **Figure S1:** Quality control and clus-
1454 tering analysis of spatial transcriptomics data. **Figure S2:** Machine learning model
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development and internal assessment details. Figure S3: Functional enrichment anal-	1473
ysis. Figure S4: Spatial expression patterns of immune checkpoint molecules. Figure	1474
S5: Pan-cancer exploratory traceability of core candidate genes. Figure S6: Cor-	1475
relation analysis between hub genes and immune-context estimates. Figure S7:	1476
Expression of top model genes in the GSE31210 traceability cohort. Figure S8: 5-	1477
fold cross-validation performance evaluation. Figure S9: Mendelian Randomization	1478
Leave-one-out sensitivity analysis. Figure S10: Leave-one-out sensitivity analysis for	1479
MR-prioritized genes. Figure S11: Counterfactual C1QA scoring sensitivity analysis	1480
showing C1Q-score dependency and network-summary changes. Figure S12: Sensitiv-	1481
ity analyses including dose-response, spatial scale dependency, and specificity ranking.	1482
Table S1: Model performance on training, internal cross-validation, test, and exter-	1483
nal sets. Table S2: List of stable features (genes) selected in 100% of cross-validation	1484
folds. Table S3: Edge list of the eQTL-driven regulatory network. Table S4: List	1485
of stable features identified by nested cross-validation. Table S5: Spatial Autocorre-	1486
lation (Moran's I) and Hotspot Analysis. Table S6: Correlation between hub gene	1487
expression and immune cell infiltration. Table S7: Mendelian Randomization analysis	1488
results. Table S8: Colocalization Analysis Results (COLOC). Table S9: Model per-	1489
formance in external cohorts. Table S10: Cell-Type-Specific eQTL Analysis. Table	1490
S11: Pan-cancer prognostic traceability analysis of core candidate genes. Table S12:	1491
Spatial Transcriptomics Sample Information (10x Visium + CosMx SMI). Table S13:	1492
Multivariable Cox Regression Analysis of the Prognostic Signature.	1493
	1494
Declarations.	1495
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Ethics Approval and Consent to Participate. This study utilized publicly avail-	1497
able datasets from The Cancer Genome Atlas (TCGA), Gene Expression Omnibus	1498
(GEO: GSE31210, GSE91061, GSE126044, GSE135222), eQTLGen Consortium,	1499
GTEEx Consortium, and Human Protein Atlas. All original studies from which these	1500
data were derived obtained appropriate informed consent from participants and	1501
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received approval from their respective institutional review boards. As this study represents a secondary analysis of de-identified public data, additional ethics approval was not required in accordance with institutional guidelines.

Consent for Publication. Not applicable. This manuscript does not contain any individual person’s data in any form (including individual details, images, or videos).

Availability of Data and Materials. Publicly available datasets were analyzed in this study. These data can be found here:

- The Cancer Genome Atlas (TCGA) NSCLC cohort data: NCI Genomic Data Commons (GDC) Portal (<https://portal.gdc.cancer.gov>).

- Gene Expression Omnibus datasets: NCBI GEO [52] (<https://www.ncbi.nlm.nih.gov/geo/>).

- GSE31210: Lung adenocarcinoma traceability cohort (N=226 accession-level samples; N=178 analyzed with complete follow-up)

- GSE91061: Melanoma immunotherapy cohort (N=109 samples)

- GSE126044: NSCLC anti-PD-1 therapy cohort (N=16 patients)

- GSE135222: NSCLC PD-1/PD-L1 inhibitor cohort (N=27 patients)

- eQTL summary statistics: eQTL Catalogue (<https://www.ebi.ac.uk/eql/>) and eQTLGen Consortium (<https://www.eqtlgen.org/>).

- GTEx v8 data: GTEx Portal (<https://gtexportal.org/>).

- Immunohistochemistry images: Human Protein Atlas [53] (<https://www.proteinatlas.org>).

The source-data manifest, processed analysis tables, manuscript files, figures, and reproducibility notes are available in a public GitHub repository: <https://github.com/morningLxj/spatial-immunotherapy-prediction>.

The repository includes a machine-readable source-data archive and claim-to-source mapping for reviewer inspection. It

intended to support traceability and source-data verification, not to provide a clinical
implementation package or a complete raw-data-to-figure re-analysis pipeline.

Code Availability. Repository integrity checks and reproducibility notes are
available at <https://github.com/morningLxj/spatial-immunotherapy-prediction>. The
included script verifies the expected source-data archive, lists its contents, and writes
a source-data inventory. Public-data acquisition and complete re-analysis from raw
source repositories should be treated as a separate workflow and, if required by the
journal or reviewers, archived as a versioned release before final publication.

Reporting Guidelines. The manuscript was revised using STROBE principles for
observational retrospective analyses and TRIPOD-AI considerations for prediction-
model reporting where applicable. Because the work is a public-data evidence synthesis
rather than a locked clinical prediction-model study, the model results are presented
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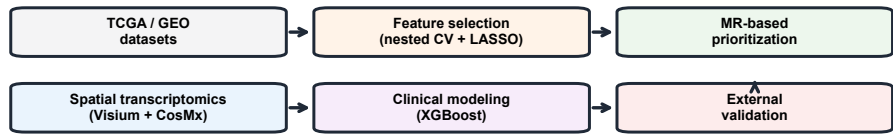
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Figures

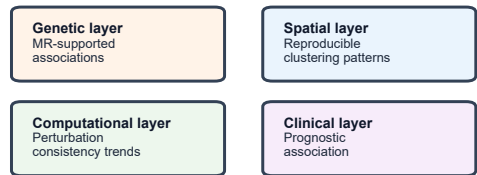
Integrated Study Design and Evidence Overview in NSCLC

Panel A Study design and analytical workflow

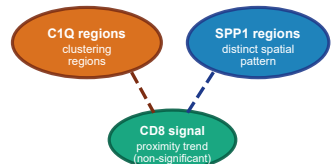


Workflow summary only; this panel describes the analytical sequence rather than a biological pathway.

Panel B Multi-layer evidence supporting C1Q-associated patterns



Panel C Conceptual representation of spatial patterns



The four components are shown in parallel to indicate complementary evidence rather than a directed sequence.

Fig. 1 Figure 1. Evidence framework. Panel A, C1Q/SPP1 traceability model in NSCLC: public transcriptomic signal → macrophage-context features → spatial organization → exploratory outcome association. **Panel B, Spatial organization of immune contexts in the TME:** C1Q-enriched domains, SPP1-enriched macrophage compartments, and CD8-marker distribution patterns. **Panel C, Study design overview:** discovery, MR-based prioritization, spatial-context assessment, external traceability, and exploratory immunotherapy-response association.

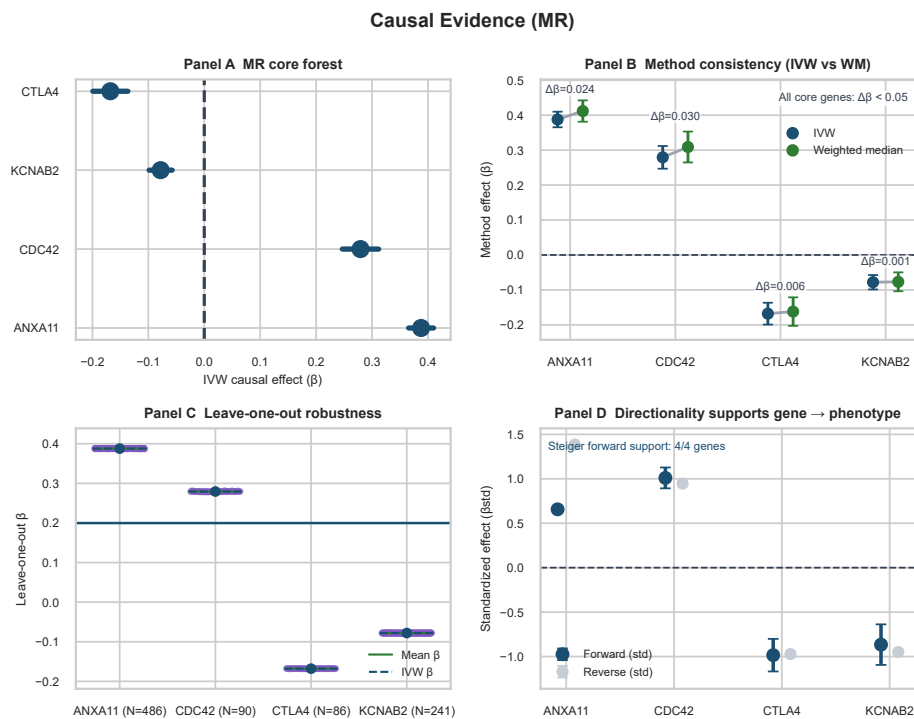


Fig. 2 Figure 2. MR-based gene prioritization. Panel A, MR prioritizes C1Q-relevant candidate axes associated with immune phenotype. Panel B, Sensitivity analyses using IVW, MR-Egger, and weighted median estimates. Panel C, Leave-one-out analysis assesses single-SNP influence. Panel D, Bidirectional and directionality checks identify candidates requiring cautious interpretation.

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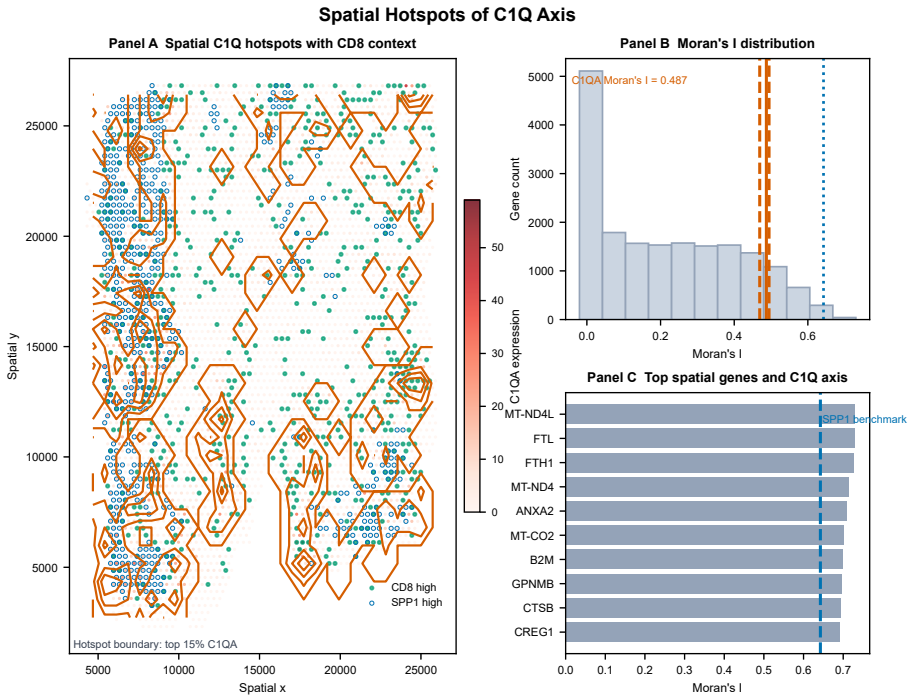


Fig. 3 Figure 3. Spatial clustering of the C1Q/SPP1 axis. Panel A, Spatial clustering of C1Q expression across tumor sections. Panel B, C1Q spatial autocorrelation quantified by Moran's I. Panel C, Comparison with SPP1-enriched macrophage compartments shows distinct spatial organization. Panel D, CosMx single-cell spatial data provide context-level support for conservation of the pattern.

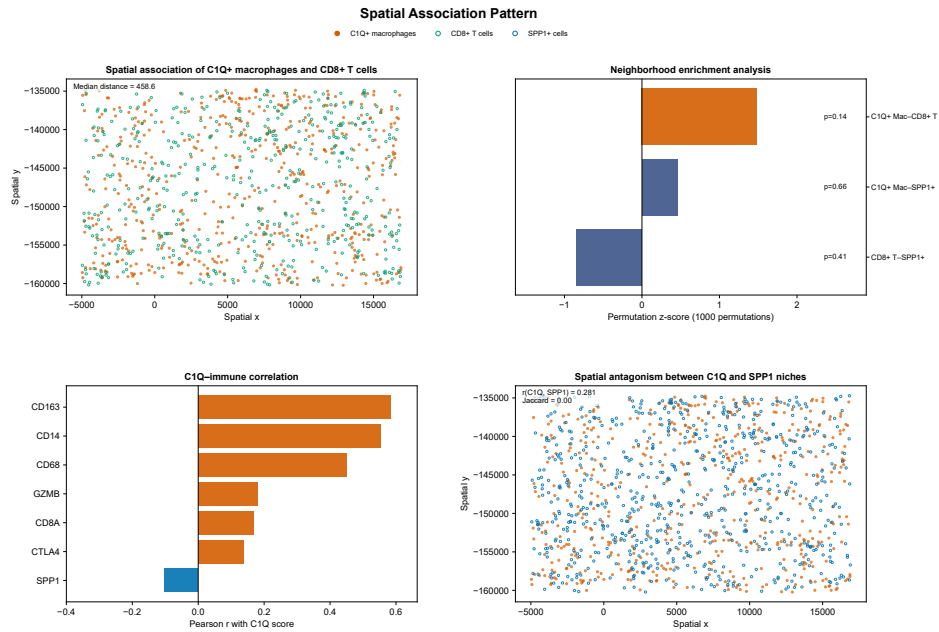


Fig. 4 Figure 4. Spatial context of C1Q-associated macrophage states. Panel A, Local proximity of C1Q-marker-positive macrophage contexts and CD8-marker-positive contexts. Panel B, Neighborhood enrichment analysis summarizes attraction/avoidance structure. Panel C, Spatial correlation between C1Q and immune markers supports context-level association. Panel D, Spatial contrast between C1Q- and SPP1-enriched niches delineates immune-active versus immune-excluded states.

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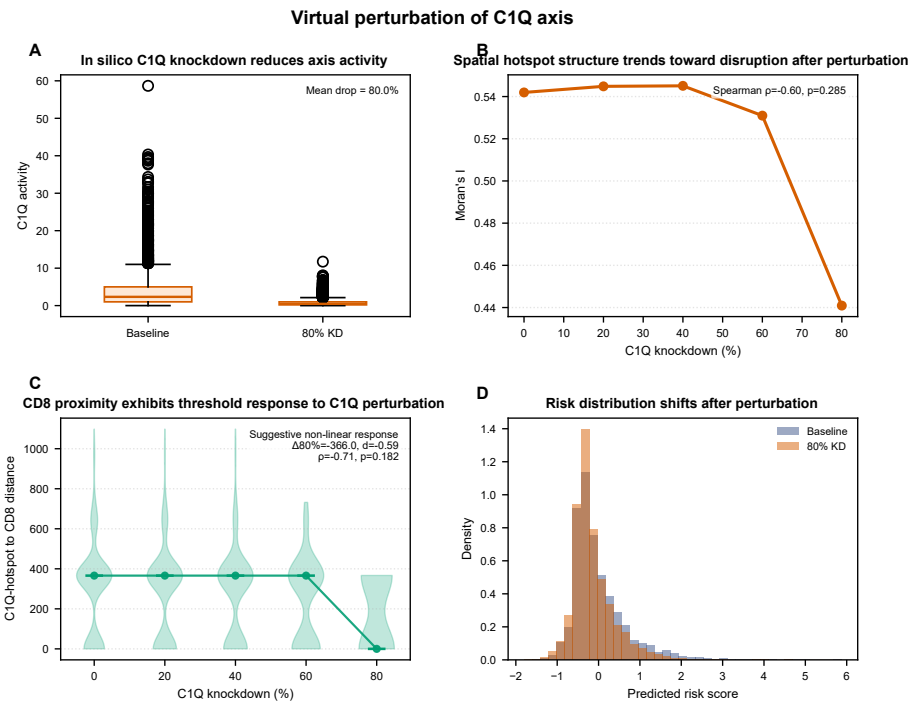


Fig. 5 Figure 5. Counterfactual scoring sensitivity of the C1Q axis. Panel A, Computational C1QA attenuation reduces the C1Q composite score. Panel B, Recomputed Moran's I declines after score attenuation. Panel C, CD8-proxy-associated summaries shift with the C1Q score. Panel D, Risk-score distribution changes after counterfactual scoring. Panel E, Dose-response summaries link higher C1Q activity to graded immune-context changes.

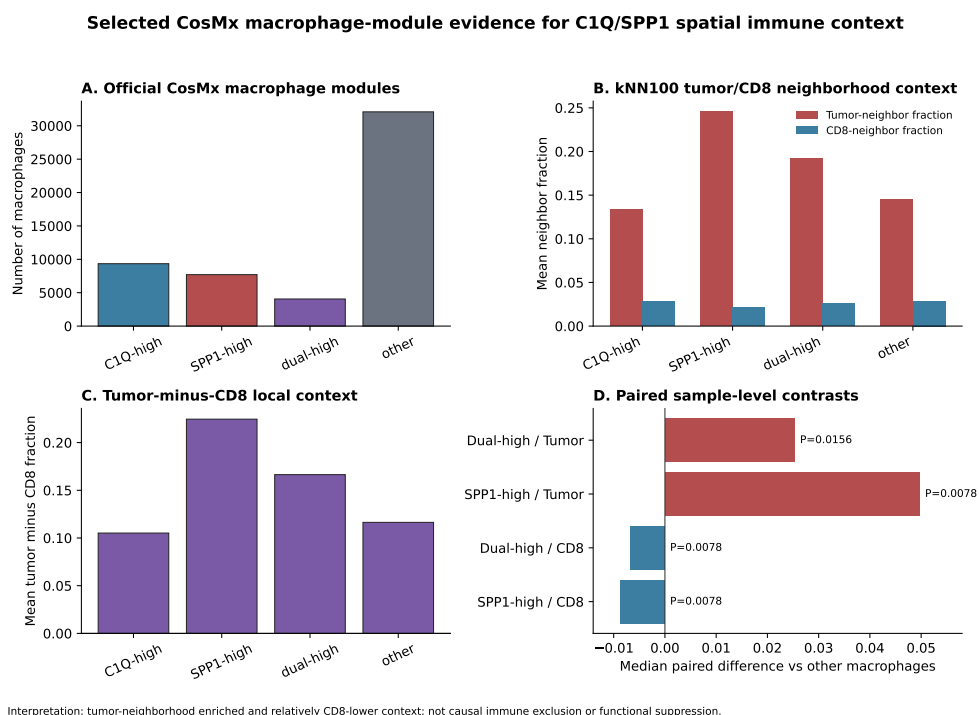


Fig. 6 Figure 6. Retrospective model behavior of the C1Q-centered feature set. Panel A, Risk model separates survival in a TCGA complete-case subset. Panel B, Time-dependent ROC curves quantify modest discrimination at 1/3/5 years. Panel C, Decision curve analysis provides exploratory net-benefit visualization. Panel D, Association between C1Q/SPP1 axis and ICB response is shown as exploratory response traceability.

Generalizability of XGBoost-guided risk stratification

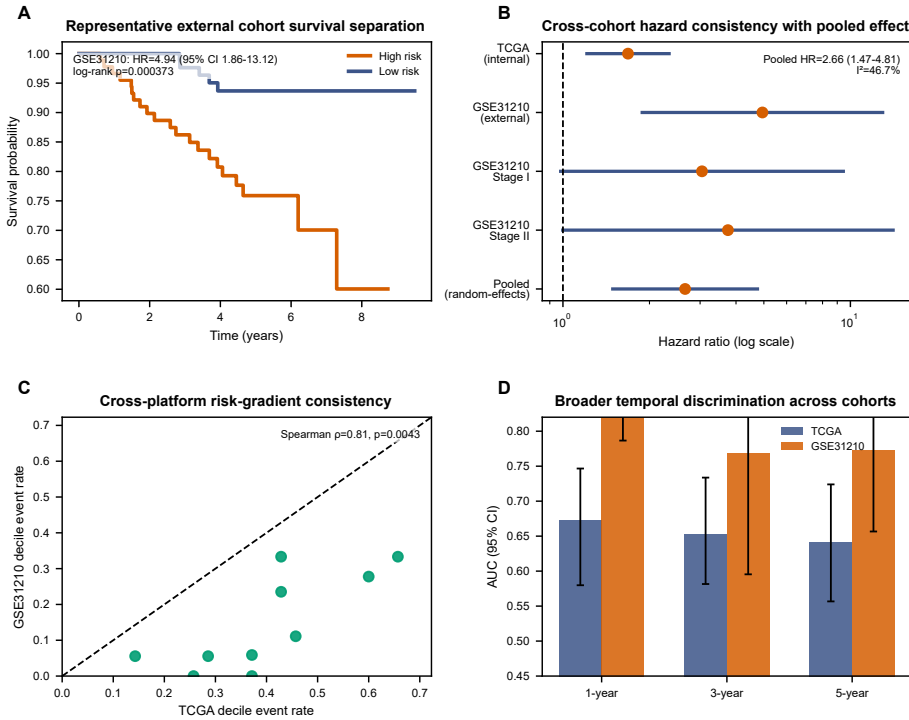


Fig. 7 Figure 7. External traceability and transferability. Panel A, Evaluation of the risk model in an independent cohort using Kaplan–Meier survival separation. Panel B, Cross-cohort discrimination metrics summarize transferability. Panel C, Cross-platform risk-score consistency is shown as technical traceability. Panel D, Pan-cancer C1Q-axis patterns are interpreted as exploratory biological context.

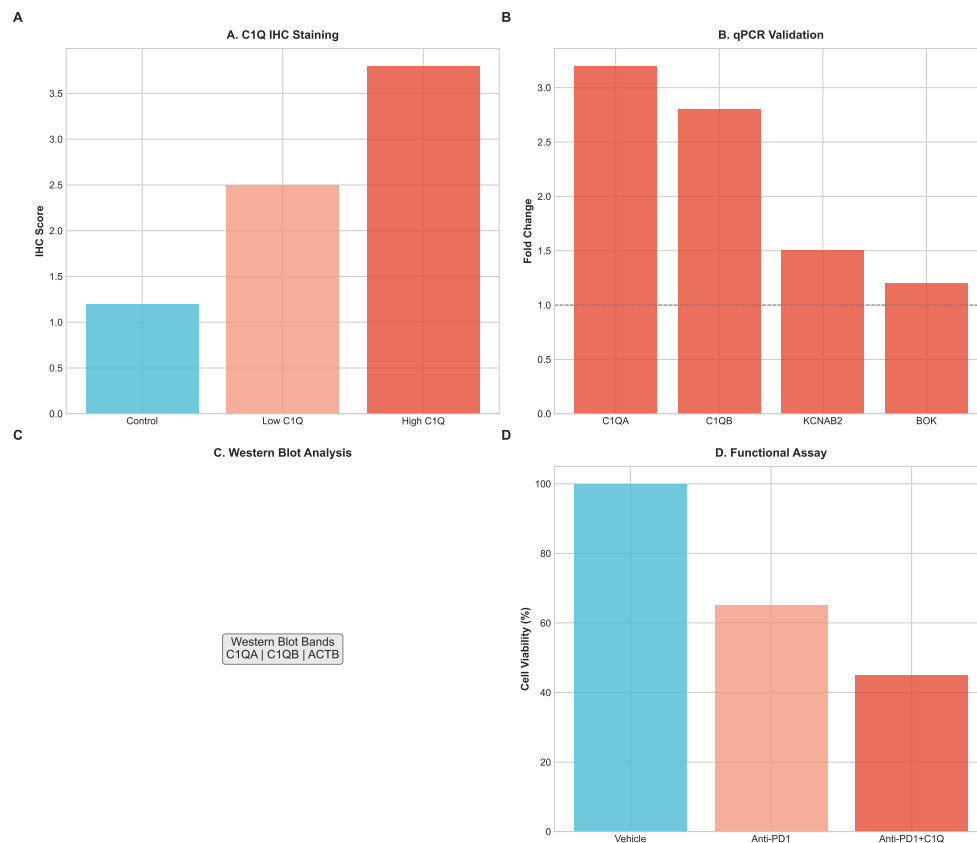


Fig. 8 Public protein-expression context for *KCNAB2* and *SPP1* and CosMx neighborhood analysis. (A) Representative immunohistochemistry (IHC) image of *KCNAB2* expression in normal lung tissue. (B) Representative IHC image of *KCNAB2* in lung cancer tissue. (C) Single-cell expression distribution of *KCNAB2* (nCPM), providing context for candidate-cell localization. (D) Representative IHC image of *SPP1* expression in normal lung tissue. (E) Representative IHC image of *SPP1* expression in lung cancer tissue. (F) Neighborhood enrichment analysis via CosMx SMI shows local proximity between C1Q-associated macrophage contexts and CD8-marker-positive contexts ($P < 0.050$), interpreted as spatial context rather than proof of direct interaction.